

Abhyuday Vaish

abhyudayvaish.work@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science (CGPA: 9.02)

Chennai, India

July 2024 – May 2028

Delhi Public School

Class X: 95.6% — Class XII: 86.8%

Lucknow, India

2022 – 2024

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (Postgres), HTML/CSS/JS, MATLAB

Frameworks: Flask, FastAPI, PyTorch, TensorFlow, LangChain, HuggingFace, Scikit-learn, OpenCV

Tools: Git, Docker, Cursor, Claude Code, Google Cloud Platform, Render, VS Code, Linux/Bash

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Linear Algebra, Probability & Statistics, Computer Networks

CERTIFICATIONS

- **Deep Learning (NPTEL – IIT Madras):** Elite + Gold Certification (Score: 90%+).
- **Machine Learning Specialization (Stanford University):** Supervised, Unsupervised & Reinforcement Learning.
- **Generative AI for Software Development (DeepLearning.AI):** Applied GenAI patterns and LLM integration.

EXPERIENCE

Samsung PRISM Intern, Samsung R&D

March 2026 – Present | Remote

- Working with small language models to support research experiments, evaluation workflows, and model behavior analysis.

Research Intern, IIT BHU

Mar 2026 – Apr 2026 | Remote

- Supporting biomedical research on deep learning methods through data preprocessing, model experimentation, and result analysis.

AI/ML Development Intern, Daira EdTech

Aug 2025 – Oct 2025 | Remote

- Automated a pipeline ingesting 100k+ UPSC PYQs, using OpenCV (adaptive thresholding, CLAHE) for high-accuracy text extraction from scanned documents.
- Launched a RAG/Generative AI system transforming static questions into dynamic assessments with auto-generated explanations, driving 40% user engagement growth.

Summer Research Intern, Vellore Institute of Technology

May 2025 – July 2025 | Chennai, India

- Pioneered opponent profiling by integrating Stockfish with Chess.com APIs, achieving 89.5% recall in identifying opening repertoire vulnerabilities.
- Fine-tuned a Gemma-2B LLM to generate automated counter-strategies, achieving 81.6% validation accuracy.
- Analyzed 2,500+ games through a custom pipeline to distill exploitable tactical patterns and vulnerabilities.

PROJECTS

Solar CME Detection System | Python, Flask, TensorFlow, PyTorch, Keras, PostgreSQL

Jan 2026 – May 2026

- Developed a stacked LSTM pipeline forecasting Kp-index 1h ahead from 6h solar wind windows, benchmarked against PINN and Quantum architectures on 14,700+ samples (± 1 Kp tolerance).
- Integrated an ETL pipeline ingesting 20 months of Aditya-L1 ASPEX telemetry alongside real-time DSCOVR/NASA feeds into a PostgreSQL datastore.
- Deployed a Flask dashboard with 3-tier RBAC, real-time CME shock alerts via statistical anomaly detection, and scientist-restricted on-demand model retraining.

Traffic Violation Detection System | Python, YOLOv8, PyTorch, OpenCV, ByteTrack, EasyOCR, Streamlit

June 2026

- Fine-tuned custom YOLOv8 models on a 6,500-image dataset for helmet, seatbelt, and license plate detection using adaptive image preprocessing.
- Applied ByteTrack-based trajectory analysis and geometric heuristics to identify 6 traffic violations in real time.
- Delivered an end-to-end enforcement pipeline integrating EasyOCR, SQLite, ReportLab, and Streamlit for automated e-challan generation and analytics.

AI Academic Assistant | FastAPI, Next.js, Supabase, Playwright, Tesseract OCR, pgvector

June 2025 – May 2026

- Architected an AI academic assistant indexing 850+ PYQs across 170 courses with 5 context-aware chat modes for guided exam prep.
- Created RAG/OCR pipelines for automated note summarization and syllabus-aligned quizzes, saving students ~5 hours of manual revision per exam.
- Automated PYQ-trend-driven question paper generation across 8 syllabus modules to help students target high-weightage topics.
- Containerized all services with Docker and automated build-test-deploy workflows using GitHub Actions CI/CD, enabling seamless deployments to Render with zero manual intervention.

AWARDS & ACHIEVEMENTS

- **1st Place**, Industry-Academia Innovators Conclave (IAIC 2025): Research Poster Presentation Category.
- **Global Rank 754** in CodeChef Starters 224 (Max Rating 1450+).
- **Max Ratings:** LeetCode (1730+), Codeforces (1330+).

POSITIONS OF RESPONSIBILITY

Project Lead | AI/ML Team Member, Microsoft Innovations Club, VIT Chennai

Sep 2024 – Present | Chennai, India

- **Project Lead (Jan 2026 – Present):** Leading a 7-member team on the “AI Academic Assistant”; conducting code reviews, overseeing deployment, and mentoring juniors on GenAI.

- **AI/ML Member:** Organizing workshops and collaborating with the AI/ML team on Kaggle competitions and technical projects.

Data Science Member, Google Developer Groups (GDG), VIT Chennai

Sep 2025 – Present | Chennai, India

- Collaborating on data science projects and exploring new technologies with the community.